Contract Studio

PDF contract editor and e-signing, rebuilt without Fabric.js

Angular 22 · pdf.js 6 · 18 field types · 5,000 pages · 200,000 fields · English / العربية

Repo: github.com/oriebalzyuot96/contract-studio
Video: docs/contract-studio-demo.mp4 (3 min 30 s)

Prepared by Orieb Ghassab · October 2026

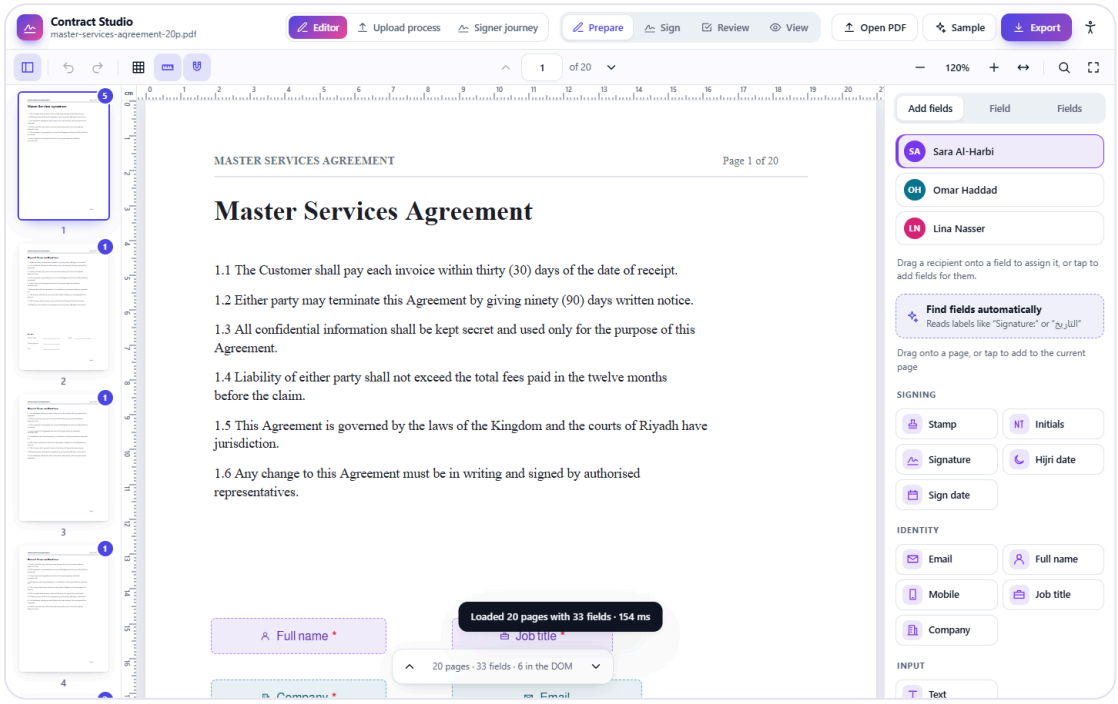
1. Why a new version

The old editor drew a Fabric.js canvas on top of every PDF page. Each canvas holds every field object in memory and redraws on each change, so large contracts get slow and heavy, and fields could end up on top of each other.

The new editor keeps the same features but draws fields as normal page elements on top of pdf.js pages. pdf.js only renders the pages near the screen, and only fields on those pages exist in the page at any time.

Topic	Old (Fabric canvas per page)	New (Contract Studio)
Large files	Slow and heavy with hundreds of pages	1,500 pages open in ~5 s; 5,000 pages in ~6.7 s
Many fields	Every field is a canvas object in memory	200,000 fields: jump ~0.1 s, nudge ~73 ms, typing ~48 ms; ~120 fields in the page
Overlaps	Fields can be placed on top of each other	Never: add, drag, resize, paste, font size all find a free spot (stress test: 0 overlaps)
Phones	Sideways scrolling, blurry pages	No sideways scroll, pinch zoom, sharp pages at the phone's pixel density
Accessibility	Canvas content is invisible to screen readers	Real buttons and inputs; axe-core: 0 violations on 6 screens

2. Editor

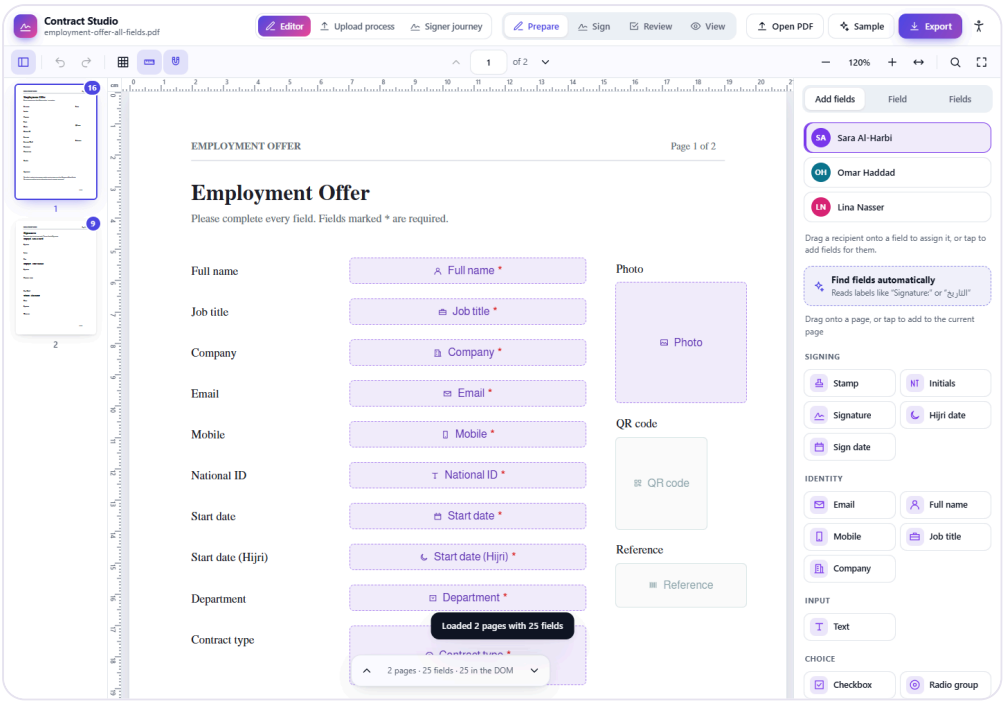


Rulers in centimetres follow zoom and scroll, shade the selected fields, and track the pointer.

- 17 field types: signature, initials, stamp, dates (incl. Hijri), name, job title, company, email, mobile, text, checkbox, radio, dropdown, image, QR, barcode…
- Drag fields from the palette, or drag a signer onto a field to assign it.
- Live drag preview turns red when blocked and shows where the field will land.
- Undo / redo, copy / paste, duplicate, align, distribute, grid and snapping, keyboard nudging.

3. All 18 field types

Try them all with **Sample → "2-page sample with every field type"**: an employment offer whose fields are spread over three signers (each signer has a colour).

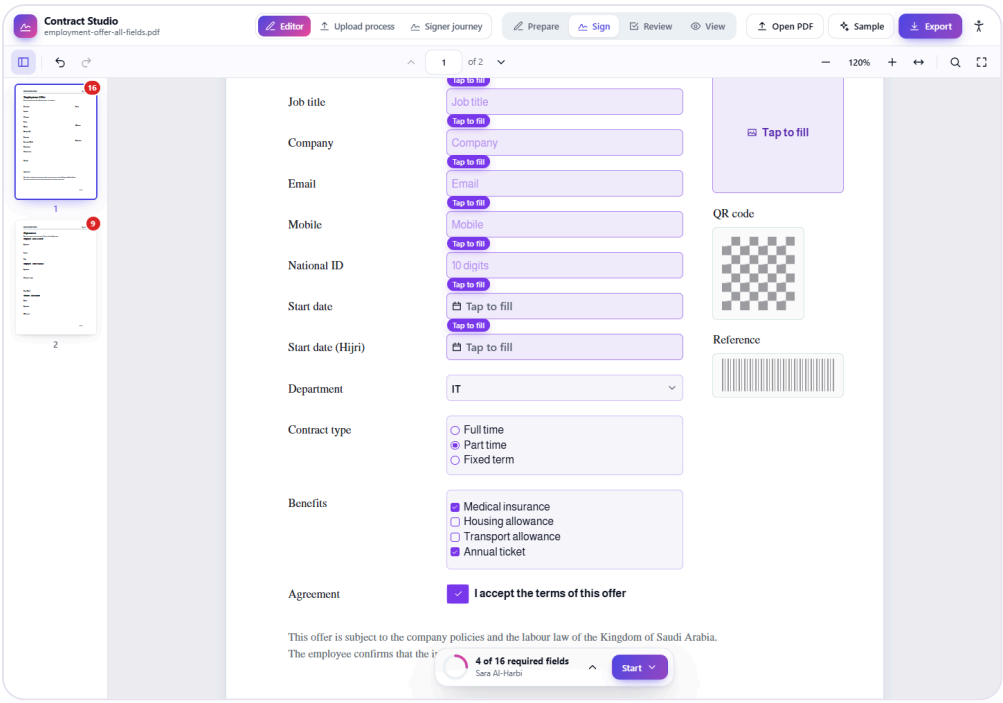


Field	What the signer does	Settings
Signature	Type (signature fonts), draw, or upload an image	Ink colours, default signature, force apply to every copy
Initials	Same as signature, short	Default initials, force apply; often placed on every page
Stamp	Upload or pick a department stamp	Round, square or rectangle; department
Sign date	Custom date picker (Gregorian)	Date format; filled automatically when signing
Hijri date	Custom date picker (Umm al-Qura)	Shows both calendars
Full name · Job title · Company	Type once	Force apply fills every copy; default value; character limit
Email · Mobile	Type; checked as you type	Email format; Saudi mobile (05xxxxxxxx)
Text	Free text	Hint, character limit, validation (number, National ID, IBAN…), text direction
Dropdown	Pick one from a custom list (keyboard and type-ahead)	Options, default value
Radio group	Pick exactly one	Options
Multi-select New	Tick several choices	Options; saved in list order
Checkbox	Tick to agree	Label beside it, checked by default, required
Image	Upload a photo	Optional / required

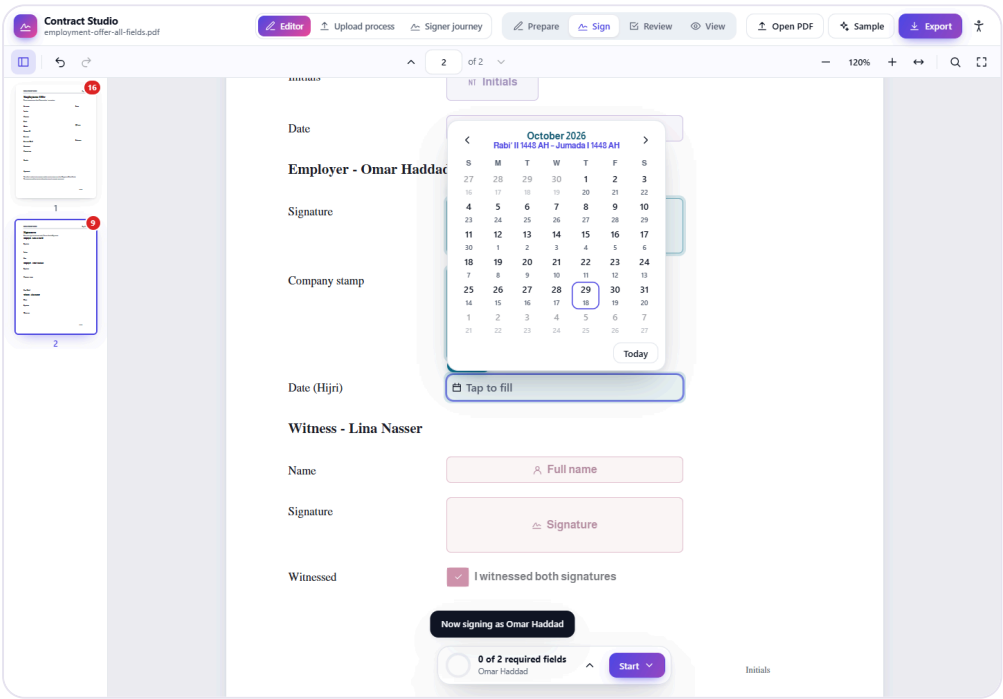
QR code · Barcode	Read-only reference on the document	Value

Every field also has: signer, required, read-only, lock position, fixed size, font, size, bold / italic / underline, colours, alignment, and placement on all / first / last / custom pages.

Filling every field in Sign mode

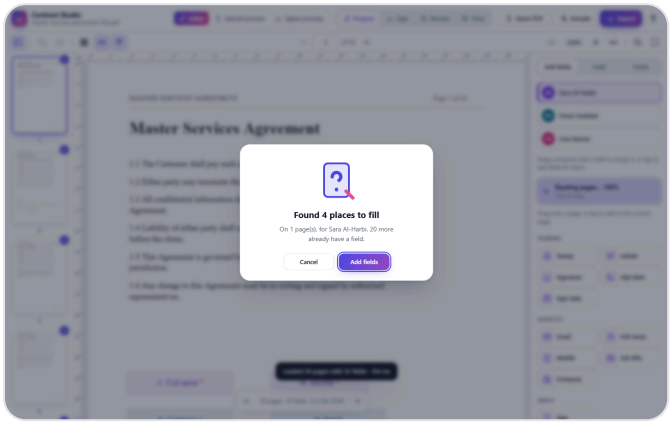


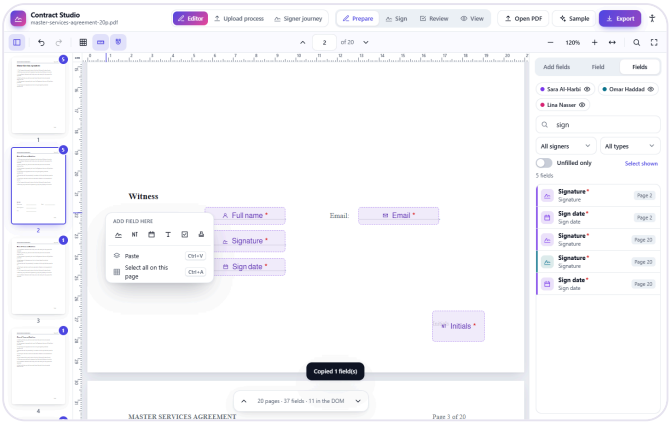
Every field is clickable: text boxes, date pickers, dropdown, radio buttons, multi-select and checkbox. **Tapping another signer's field** switches to that signer ("Now signing as …") and fills it in the same tap. The checkbox ticks, and a date picker or list opens.



4. New editor features

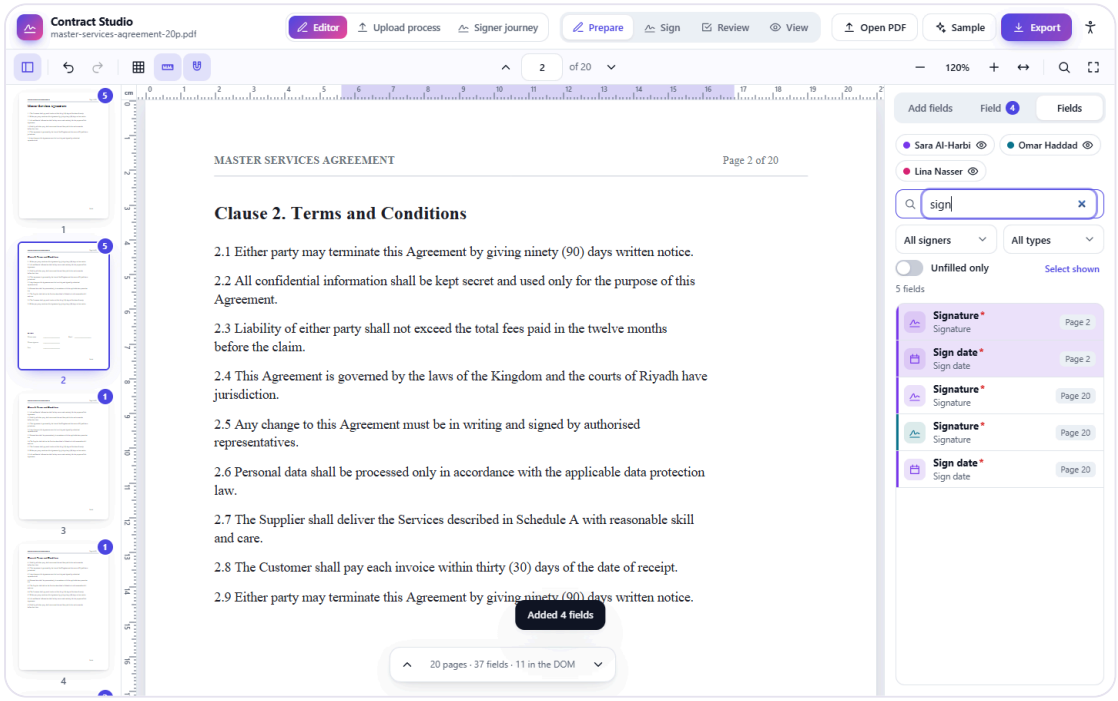
New Find fields automatically





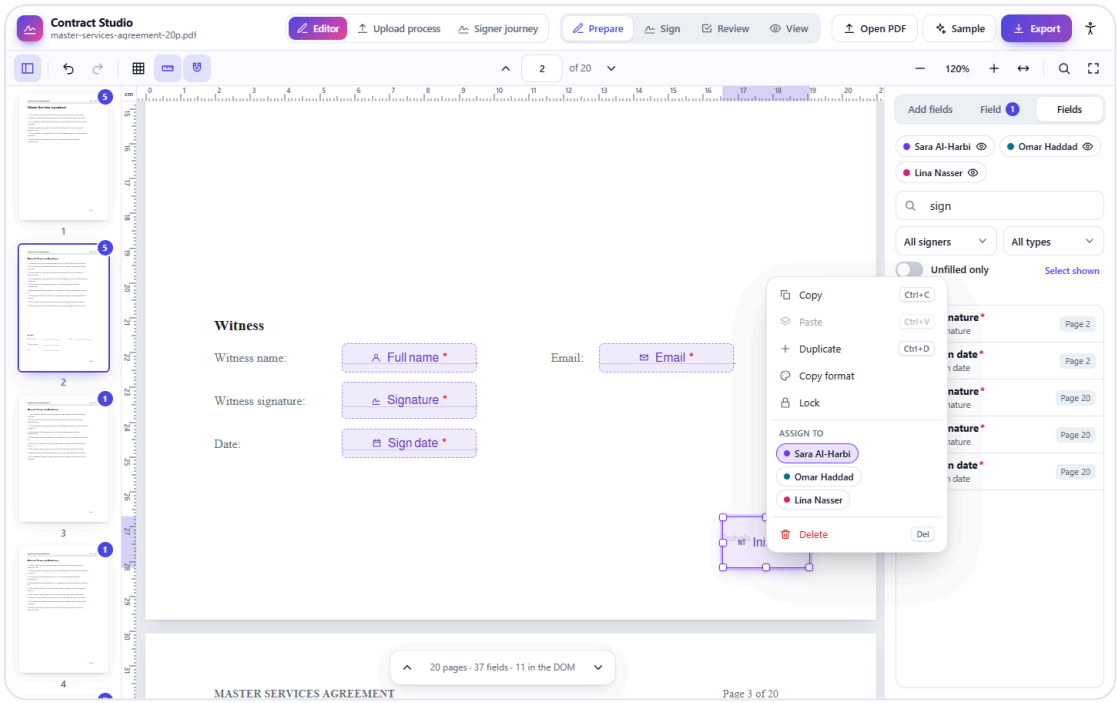
Reads the PDF text for labels such as “Signature:”, “Date:”, “Name:”, “التاريخ” ,”التوقيع” next to a blank line, and adds a field on that blank line. Places that already have a field are skipped. Shows progress and can be stopped.

New Fields list



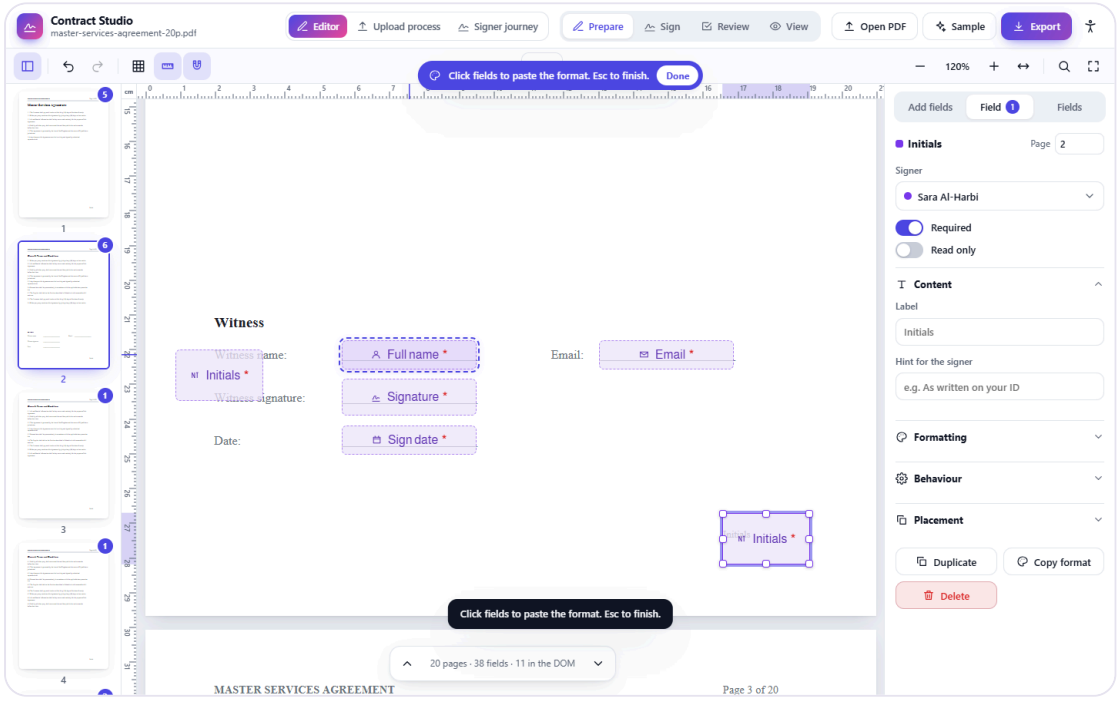
Every field in reading order. Filter by text, signer, type, or unfilled only. Click to jump; Ctrl/Shift-click to add to the selection; “Select shown” selects all results. Hide a signer’s fields from view (they still block placement). The list only draws the rows on screen, so it stays fast with 200,000 fields.

New Right-click menu



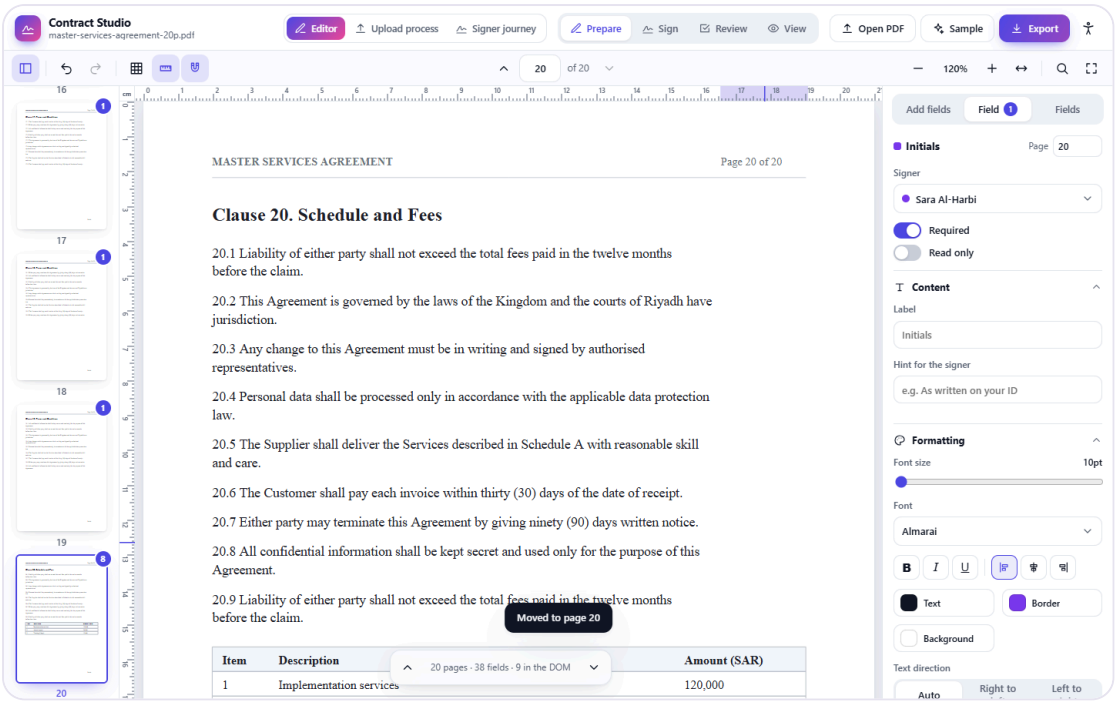
On a field: copy, paste, duplicate, copy format, lock, assign to a signer, align, delete. On an empty spot: quick-add a field or paste exactly there. Opens from the keyboard with the Menu key or Shift+F10; arrow keys move between items.

New Format painter



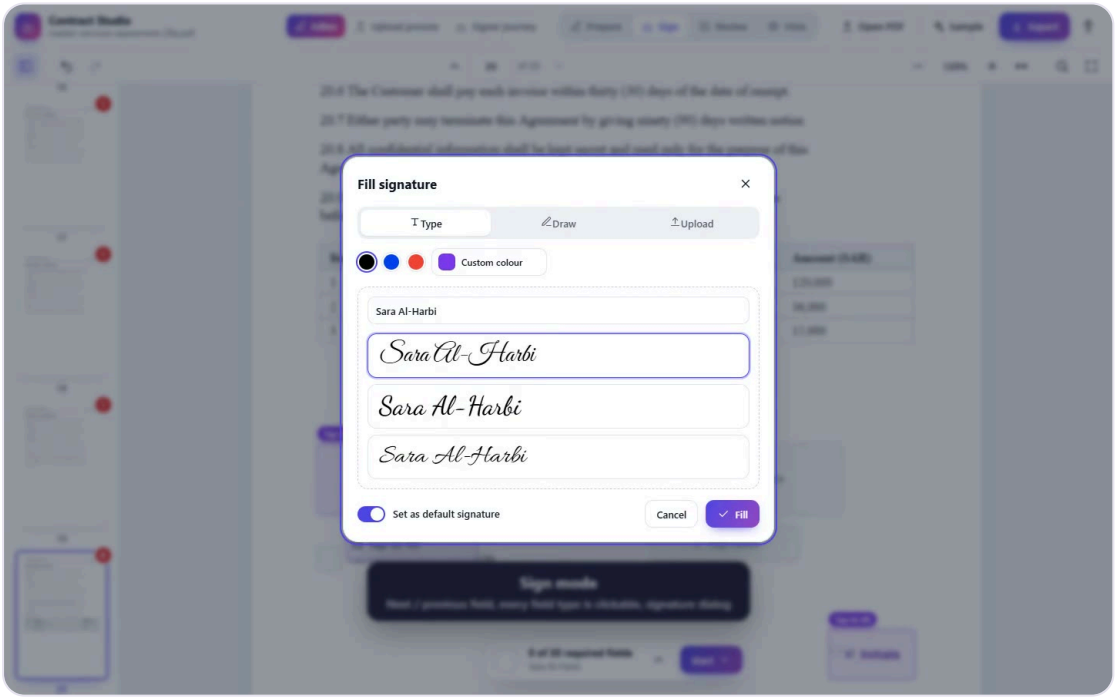
Copy font, colours, size and text direction from one field and click other fields to apply. A bar at the top shows it's on; Esc or a click on an empty spot ends it.

New Field placement: first page / last page



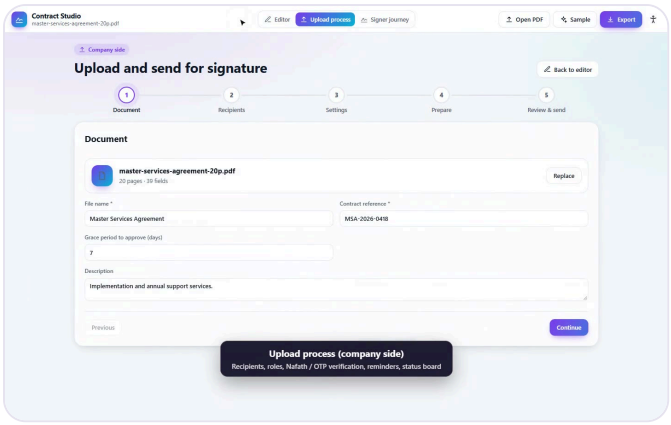
Place a field on all pages, the first page, the last page, or a custom range like “1-3, 5”. First and last page are available for every field type.

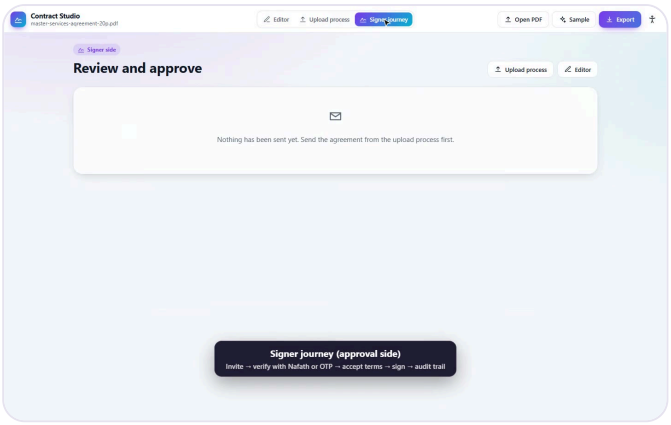
5. Signing



Sign mode moves through the signer’s fields with next / previous. Every field is clickable: inputs, checkboxes, dropdowns, Gregorian and Hijri date pickers. The signature dialog supports type, draw and upload, ink colours with a custom colour picker, “set as default”, and force-apply to all the signer’s fields.

6. Upload and approval processes (Saudi features)





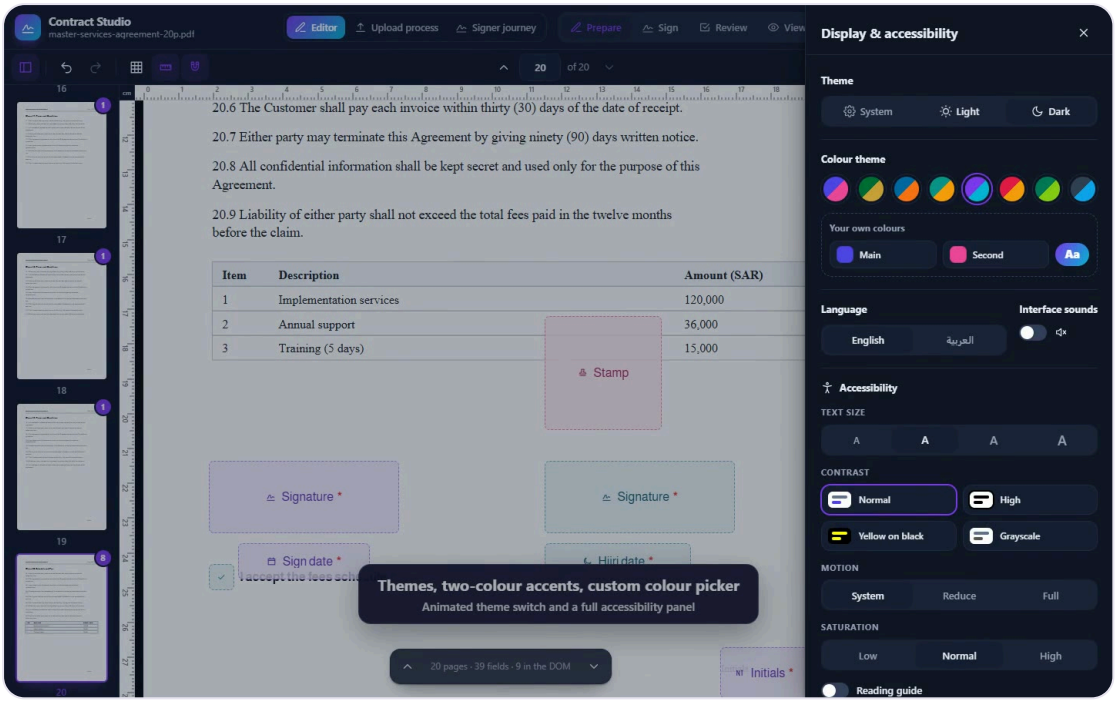
Upload process (company side): document details, recipients with roles and order, verification per recipient (Nafath, SMS OTP, authenticator), reminders, expiry and grace period, then a live status board.

Signer journey (approval side): invitation → identity check with Nafath or OTP → accept terms → sign → done, with a full audit trail and a SHA-256 hash of the document.

7. Search and field navigation

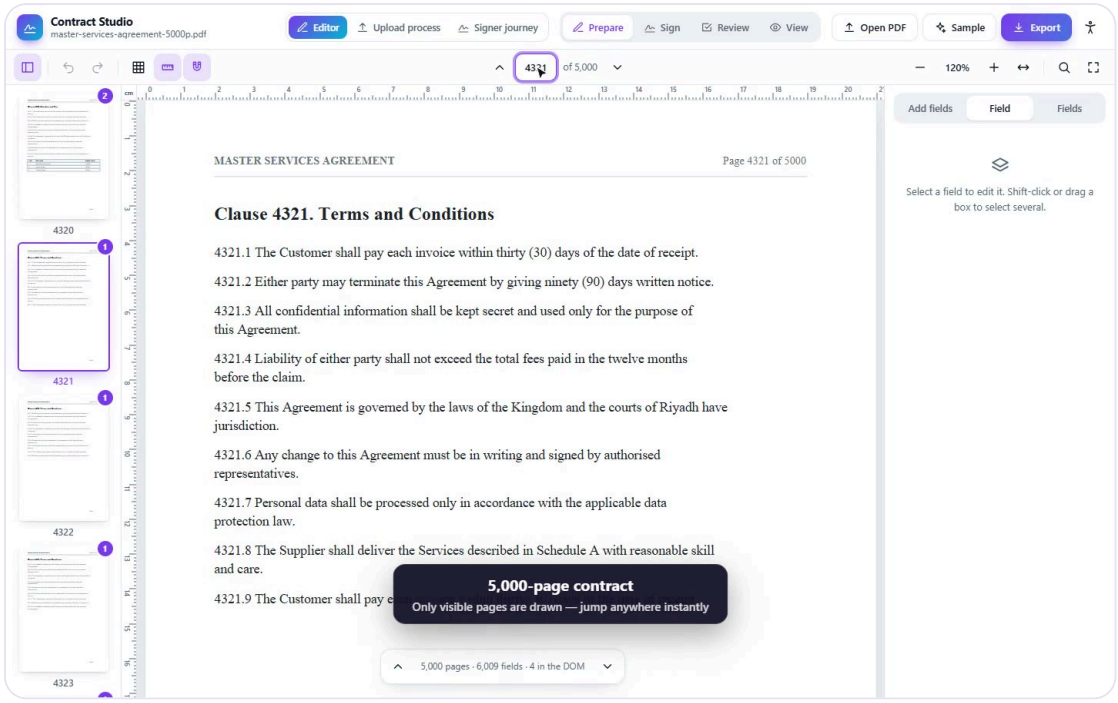
Search inside the PDF text with whole word, match case and diacritics options, hits per page, and matching fields. Voice search uses the browser's speech recognition. Next / previous field works in every mode: it keeps your zoom and scrolls up/down *and* left/right so the field is always fully on screen.

8. Themes, sounds and accessibility

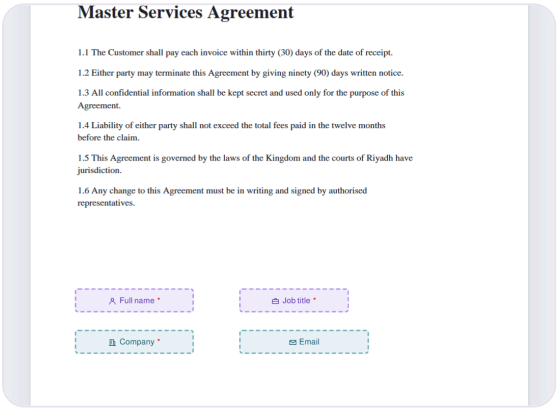


- Light / dark / system with an animated circular theme switch; two-colour accent themes and a custom colour picker (no browser colour input).
- Accessibility panel: text size, high contrast, yellow-on-black, grayscale, reduced motion, saturation, readable font, spacing, big cursor, strong focus, underlined links, reading guide, read-aloud on focus. “Reset accessibility” and “Reset all settings”.
- Interface sounds (can be turned off), custom loader, illustrated alert dialogs, three.js start scene.
- Full English and Arabic (right-to-left) interface.

9. Big documents and phones



A generated 5,000-page contract, jumped straight to page 4,321.



- The page moves like paper: swipe in any direction, even starting on a field; tap selects; press and hold drags a field.
- Pinch zoom is done by the app, not the browser, so the page is redrawn sharp instead of stretched.
- Pages are drawn at the phone's pixel density (3.5× on a high-end Android). When zoomed in to 170%, Android pages stay at 2.3× (pdf.js's phone default gave 0.7×).
- PDF text uses pdf.js's own fonts, so it looks the same on every phone (Android has no Times font).
- Bottom dock and sheets for fields, field settings and the fields list.

10. Code and delivery

- Angular 22 standalone components with signals and OnPush; enums instead of string unions; SCSS with shared variables and mixins; each component in its own .ts / .html / .scss files.
- Docker image (nginx), GitHub Actions for GitHub Pages and the container registry.
- Export: fields as JSON, or a flattened PDF with the filled values.

Numbers were measured on a desktop Chrome with the generated sample contracts.